BUAD 5272: Database Management

Virginia Drug Overdose Death Dashboard

[redacted] • [redacted] • [redacted] • Pranav Prathap

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1. Introduction:

The state of Virginia aggregates drug overdose death information across counties and demographics. This information is useful because it provides an indicator for what areas and people may be more vulnerable to certain drug classes than others. This would allow for the development of public policy initiatives to address these areas of increased vulnerability. However, as it is, the data can be difficult to navigate. By default, we have death counts by geography and demography. However, we wanted to improve functionality by giving the user the ability to summarize over certain aspects of counties or drug schedules, so we also brought in county unemployment information. To account for drug schedules, we had to manually create the dataset, classifying drug classes into their legal schedules.

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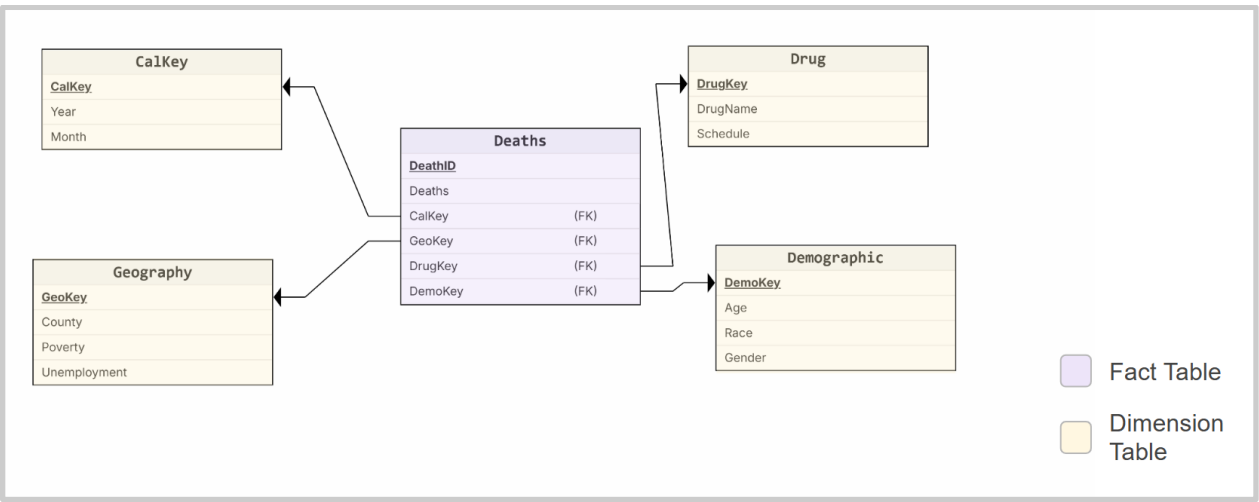
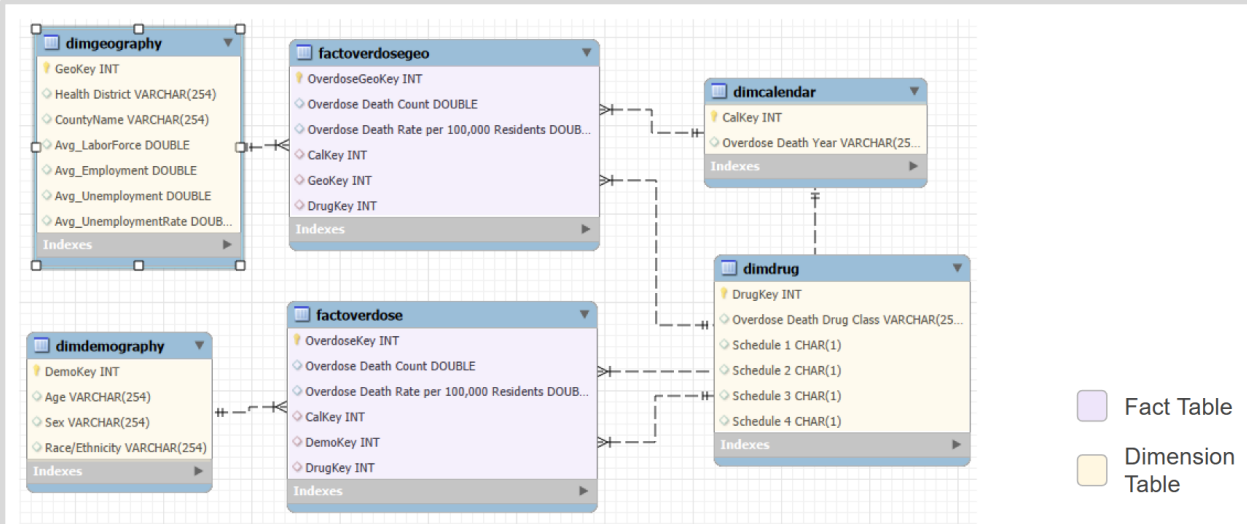
2. Initial and Final Dimensional Models:

Initial Model:



Final Schema:



Differences between the original and final model stem from issues we ran into during the ETL process, which will be discussed in the next section.

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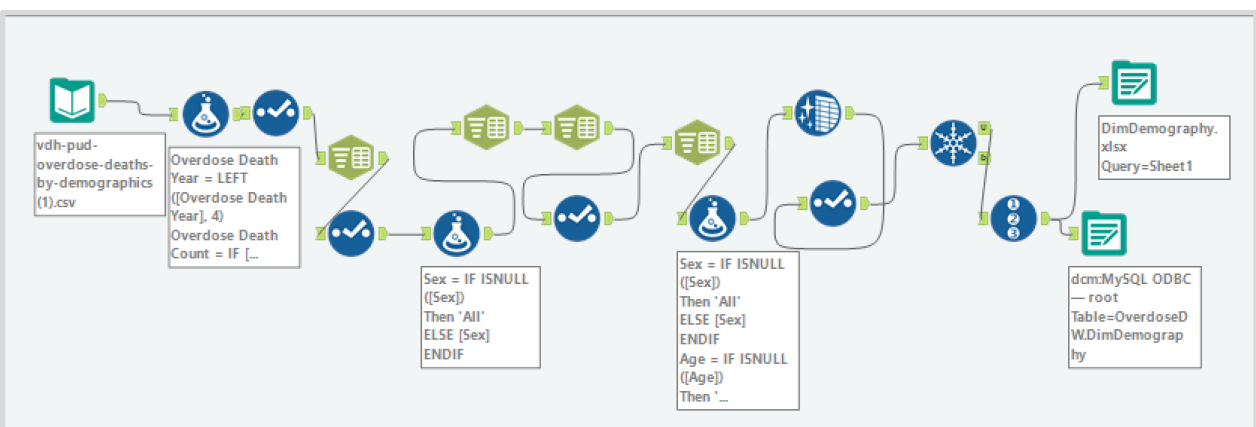
3. ETL Process:

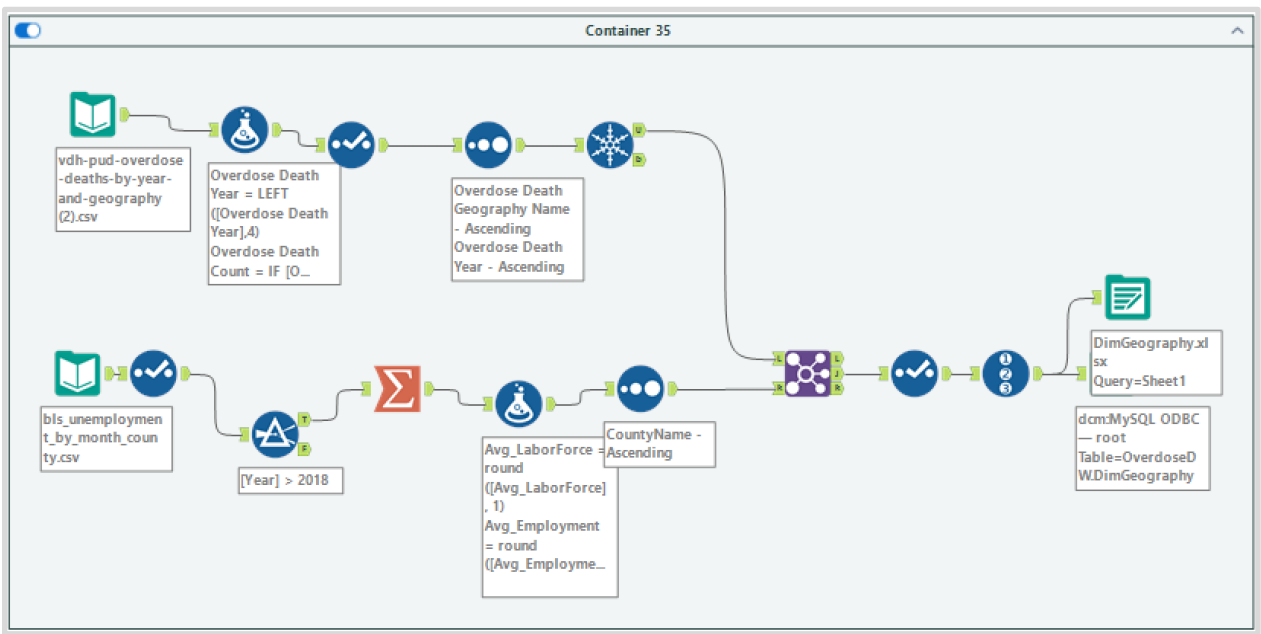
The first dimensional table we created was the demographics table with the workflow below:



The primary challenge was parsing the age, race, and sex information, which was contained in one column, into three columns. We also had to account for the fact that we don't have all combinations of all demographics. Whenever race is specified, age and sex are classified as 'all'. This does not impede the creation of the dimension, but it is

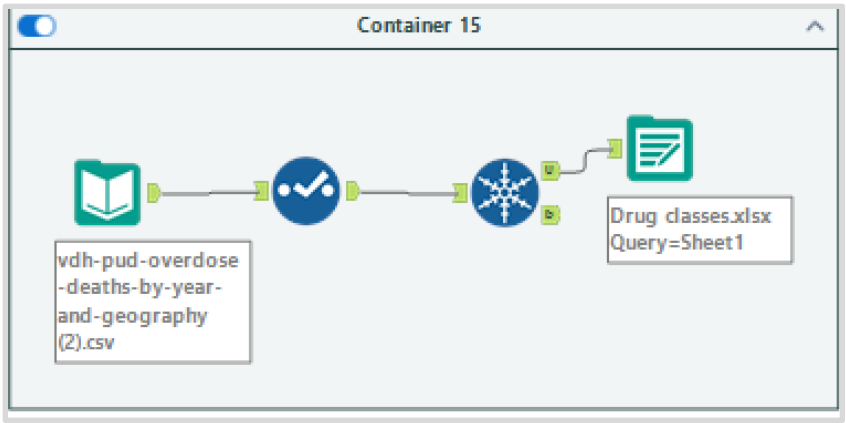
something we must be aware of when using the warehouse. We also noted at this point that because the geographic and demographic death counts are completely separate datasets, we do not have the granularity to include them both as keys in one fact table. We accounted for this by making two fact tables in our final schema, one including demographics as a dimension and another including geography as a dimension.

The second-dimensional table we created was the geography dimension using the workflow below:

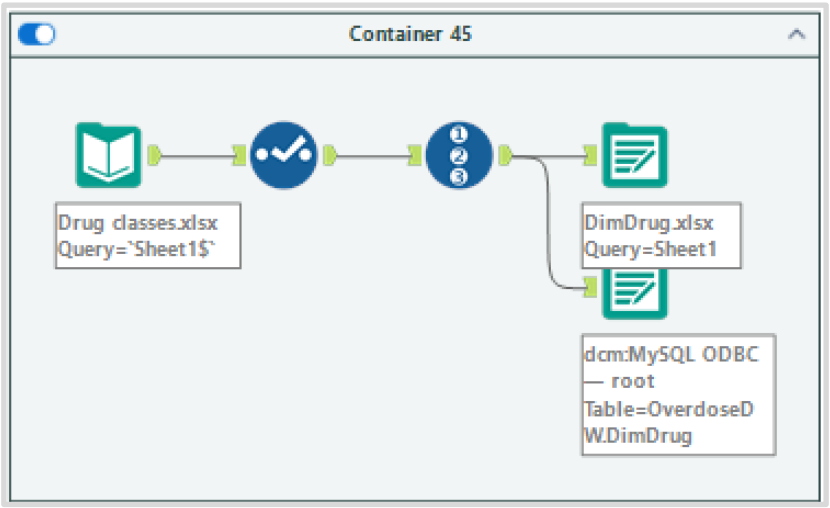


This workflow takes all unique counties used in the original county death count set and then joins them with unemployment information for said county. There are two deviations from the original schema. The first is that we could not find useful poverty information in our research, so we had to compensate by including more metrics from the unemployment data involving employment and unemployment magnitude, rather than

just the rate. The second is that because we have unemployment data for multiple years, we had to average the statistics over the relevant years (2019-2025) to use as a static value to define each geography. We could have instead converted our schema into a snowflake schema, making the geography dimension also dependent on the calendar, but we decided to keep things simplistic instead.



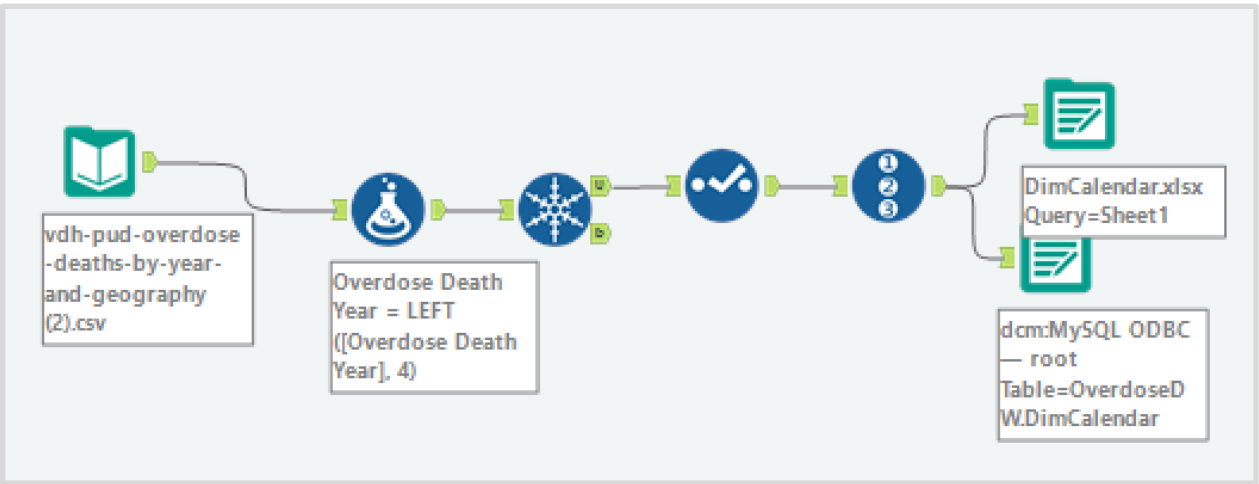
The third-dimensional table created was the drug class using the two workflows below:



The first series of steps takes all the unique drug classes found in the death count data and writes them to Excel for manual schedule classification. The second series simply takes that Excel file, applies a key to it, and then adds it to the schema. During the process

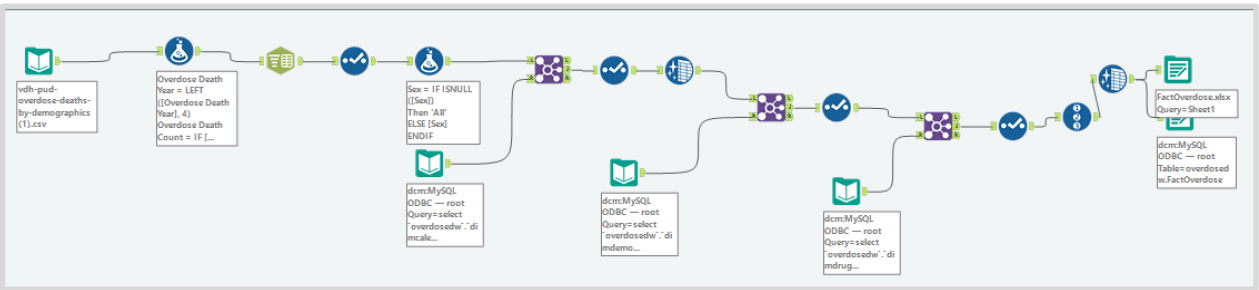
of classification, we realized that some drug classes can fall into multiple schedules, so we modified the schema to include an attribute for each schedule, allowing us to mark all the schedules that a drug class may fall into.

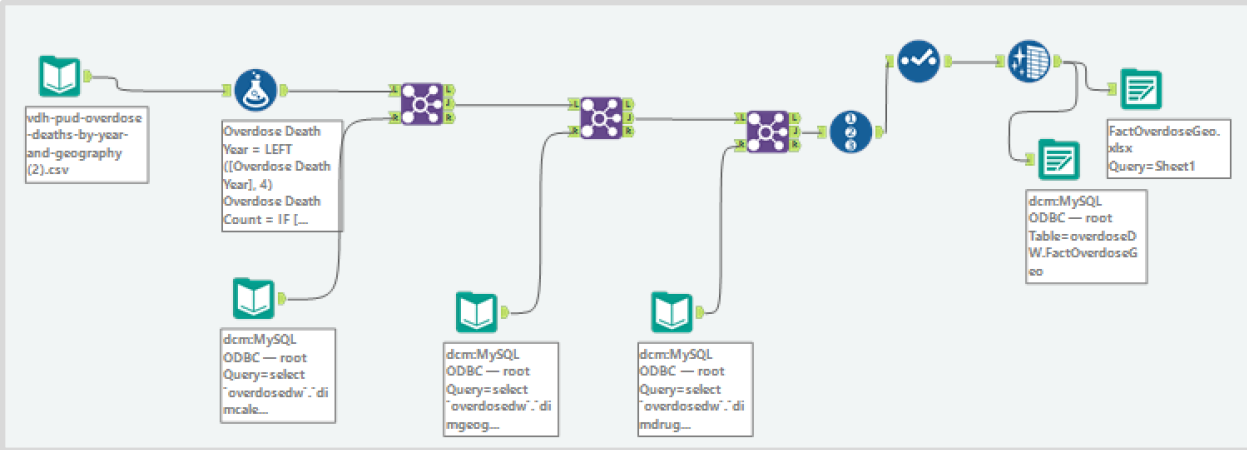
The final dimensional table created was the Calendar dimension using the workflow below:



The only issue we ran into here was that overdose death counts are only taken annually. This means that we have to reduce the granularity of the dimension and only use the year instead of the year and month.

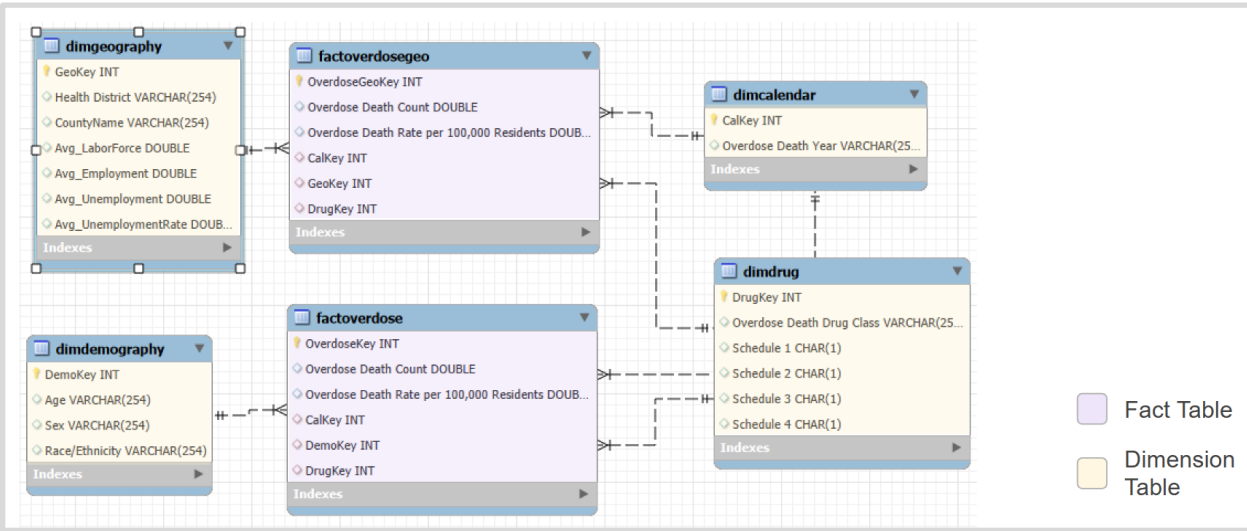
Once the dimension tables had been created, the two fact tables were created with the workflows below:





The first workflow created the table defined by demographics. The calendar dimension was joined using the year, the drug dimension was joined by the class name, and the demographic dimension was joined by the age, race, and sex. The second workflow created the table defined by geography. Calendar and drug dimensions were joined in the same way as previously, and the geography dimension was joined using the county name.

The changes made to our original design throughout the ETL process resulted in the final schema below:



The only other changes made not mentioned prior are that we also included the health district in the geography dimension for extra qualitative features to summarize over, and that we also included deaths per 100,000 residents as an additional metric in the fact table.

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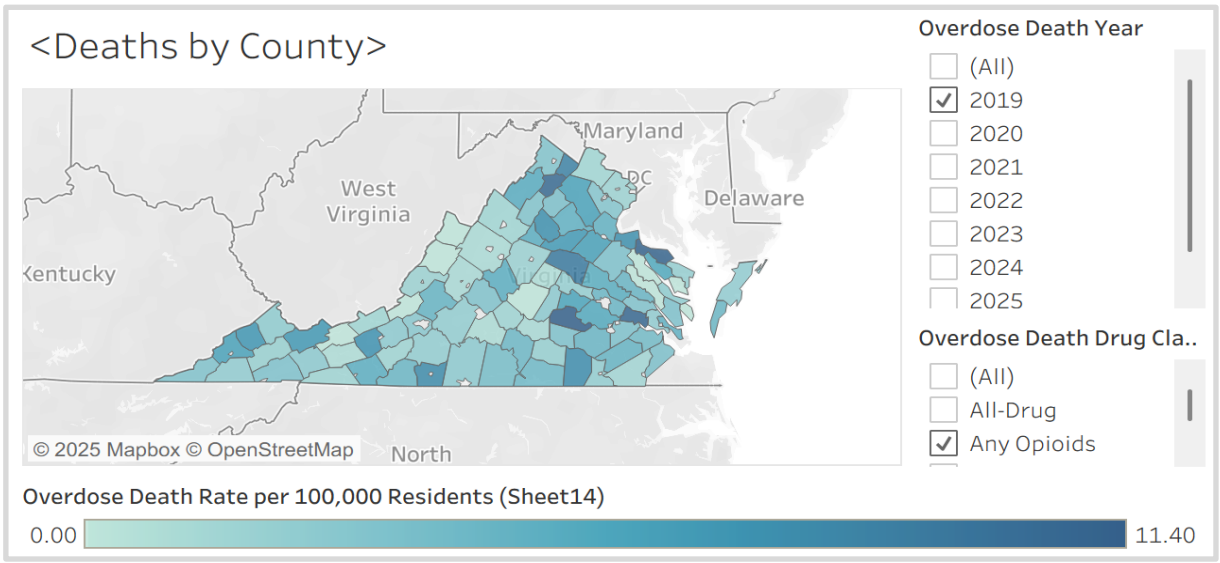
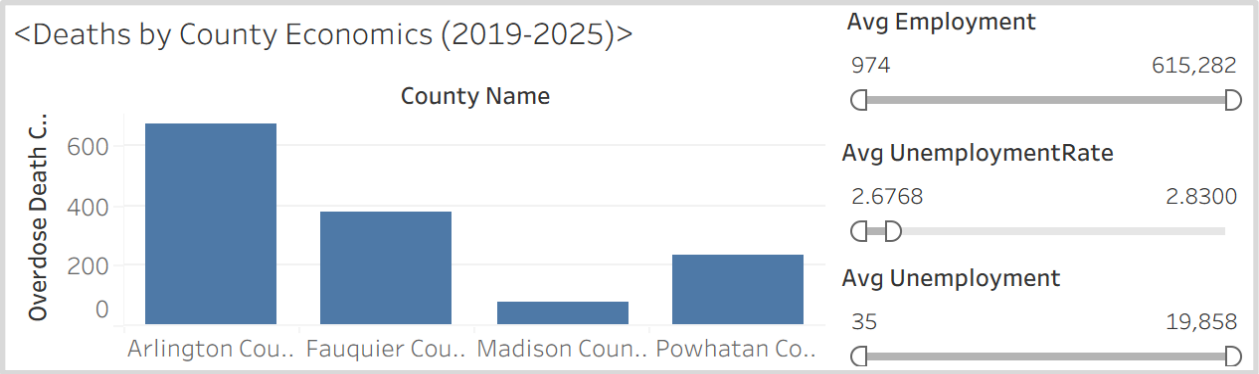
4. Analytics

Deaths per 100,000 residents by geography



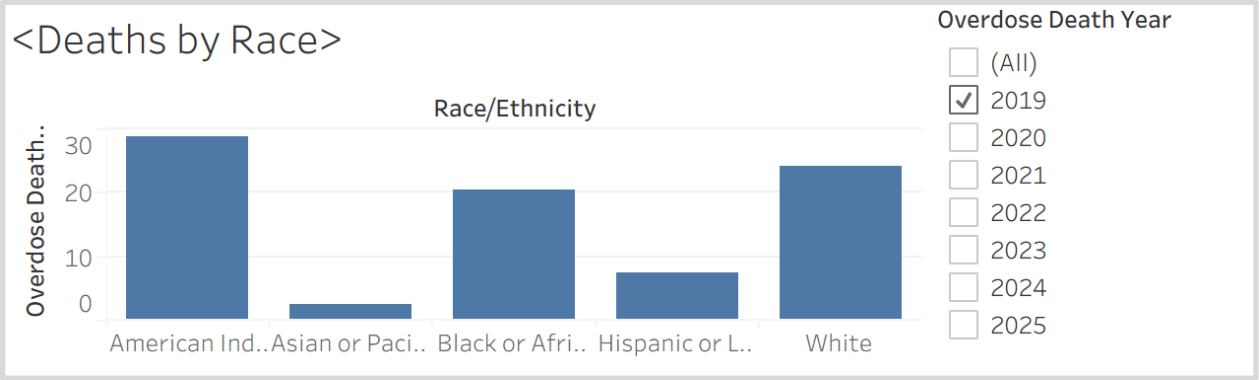
This map shows the deaths per 100,000 residents by county, with the ability to select which drugs and years the user wants to sum over. It is useful to see which counties have a higher rate of overdose death by drug

- Deaths per county by economic factors:



This graph sums the overdose deaths over the relevant period and allows the user to see the distribution of counties filtered by unemployment statistics. The user can use the rate as well as the magnitude of employed and unemployed people. When several counties are shown at once, the user can always hover the bar to see the county name when it becomes illegible on the axis.

- Deaths by Race:



This visualization allows the user to see deaths per 100,000 residents by race, also allowing them to select which years should be added to the summation. This visualization should be taken with a grain of salt because we cannot confirm with certainty whether the deaths per 100,000 residents refer to residents that fall within that racial category or if it is using a different denominator.

5. Conclusion:

The most difficult aspect of this project was working around data that is already aggregated. In our practice with data warehouses, we had the luxury of working off operational databases that had per-encounter observations. The analog to that in this case would be if our data included incident reports for each overdose death with the date, age, race, sex, and location. Instead, the data is already aggregated, meaning we had to implement workarounds such as the multiple fact tables and the limitations on how we visualize race and age demographics. That being said, there was still a surprising amount that we could do with already aggregated data, especially with the additional unemployment data that we were able to bring in. We could still reorganize it and link it to the additional data sources that allowed us to generate more compelling visualizations than had we just used it as it was. As for the visualizations themselves, we think that if we were able to do it again with more knowledge of Tableau, we could have made more visualizations to leverage the full scope of the data warehouse, such as adding visualizations that allow for choosing which drug schedules you want to visualize.

Despite the limitations of our current visualizations, we believe that the current system is in a good position for the analysis of overdose deaths within the state of Virginia. For example, the map provided can be used to observe which counties are most vulnerable to any given class of drug. Furthermore, with the data already cleaned and organized into a schema within SQL, more visualizations tailored to the end user's needs can be created very quickly. This system can also scale for the most part. The

only change we would make if this were to be adopted in the long term would be to implement the snowflake schema design so that county unemployment statistics can change from year to year. Averaging them was fine for this prototype, but if we want to scale this design as more data gets added, it would be poor design to define counties on statistics averaged over decades rather than just a few years.

It is by design that our project does not make recommendations or draw definitive inferences about trends or patterns in the dataset, as our objective was to create a tool that provides consolidated and aggregated data with supporting visualizations, which can then be used for further research. In its current state, the Virginia drug overdose death data warehouse can already be used to gain insight into which areas and people are most vulnerable to overdose from certain drug classes, and with additional time and improvements, it can become a very useful tool for addressing these social problems.